Yenatfanta Shifferaw Bayleyegn

AI Researcher | Data Scientist | Data Engineer
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LinkedIn | GitHub | Portfolio

Professional Summary

AI Researcher, Data Scientist, and Data Engineer with over 15 years of experience designing, engineering, and deploying production-grade AI systems across robotics, healthcare, blockchain, and financial technology. Experienced in scalable data engineering, machine learning, distributed software architectures, and cloud platforms, with a passion for translating cutting-edge technology research into reliable industrial solutions that create measurable real-world impact.

Core Technical Skills

- **Programming:** Python, R, SQL, Bash
- **Artificial Intelligence & Machine Learning:** Deep Learning, NLP, LLMs, Computer Vision, TensorFlow, RASA, AWS SageMaker
- **Data Engineering:** ETL, BigQuery, PostgreSQL, dbt, Trino
- **Cloud & DevOps:** AWS, Docker, Kubernetes, Prometheus, Grafana
- **Visualization:** Power BI, Google Data Studio, QuickSight, R Shiny
- **Software Engineering:** Git, REST APIs, PyTest, GoTest, CI/CD

Work Experience

AI Researcher & Developer | SingularityNET Jan 2025–Present

- Develop LLM- and ITP-based solutions for automated theorem proving and formal verification.
- Build autoformalization pipelines to transform mathematical literature into formal proofs.
- Develop AI/ML components for language technologies.
- Research AGI cognitive architectures and integrate NLP models into AI pipelines.
- Collaborate on production-ready AI systems for next-generation intelligent applications.

Senior Data Scientist | Software Engineer | EthSwitch S.C. Sep 2023–Sep 2025

- Designed and deployed ML-powered credit scoring models using financial, social media and psychometric data.
- Engineered scalable ETL pipelines and data infrastructure for analytics and ML.
- Adapted AI solutions to regulatory and privacy requirements for emerging markets.

Data Scientist | Engineer | Input Output Global Apr 2019–Nov 2024

- Developed blockchain analytics and prediction models using AWS SageMaker and Amazon QuickSight, enabling data-driven insights from decentralized systems.
- Designed interactive R Shiny dashboards integrating machine learning models to support real-time analytics and decision-making.
- Optimized dbt/Trino/BigQuery pipelines and monitoring with Prometheus/Grafana.

SME in ML & AI | Excellerent Technology Solutions Dec 2019–Jan 2024

- Built Azure GPT-based medical assistant.
- Designed and developed NLP resume parser and Android ML asset tracker.
- Provided analytics solutions for COVID-19 and farming.

Senior AI Researcher | Ethiopian Biotechnology Institute Mar 2017–Apr 2019

- Developed CNN-based kidney abnormality detection systems.
- Built deep learning solutions for healthcare and agriculture.
- Led AI training sessions and robotics initiatives.

AI Researcher | Developer | iCog Labs Jun 2014–Jan 2019

- Contributed to the development of **Sophia**, the internationally recognized humanoid robot, by implementing computer vision, sound recognition, and real-time perception systems.
- Applied OpenCog and AtomSpace/MOSES algorithms to intelligent robotic systems.

Senior Engineer | ZTE ICT Technology Sep 2011–Oct 2013

- Supported and optimized Ethio Telecom's Customer Care and Billing database, protecting sensitive data.
- Crafted query definitions for efficient data extraction and stakeholder reporting.
- Evaluated the IT infrastructure of Ethiopia's justice system and analyzed its current and future networks.

Junior Electrical Engineer | Ethio Telecom Sep 2007–Feb 2009

- Supported site surveys and equipment installation for the Fixed Line Next Generation Network.
- Helped upgrade and expand broadband services and gave customers technical support.

Education

M.Sc. Computer Engineering | Addis Ababa Institute of Technology 2008–2011
B.Sc. Electrical Engineering | Jimma University 2004–2007

Publications

- The Parent-Child Framework: Biologically Inspired Bias-Free Machine Learning Model
- Limitations of Proof of Stake Algorithm in Blockchain: A Review
- Hybrid Intelligent System for Lung Cancer Type Identification

Certifications & Professional Training

- Microsoft Certified: Azure AI Fundamentals
- AWS Certified Cloud Practitioner
- Prompt Engineering
- DevOps Fundamentals
- Kubernetes
- Database Systems
- Leadership Development

Professional Competencies

- Project Management & Agile Delivery
- Cross-functional Collaboration & Stakeholder Communication
- Mentorship, Coaching & Technical Knowledge Sharing
- Collaboration Tools: Jira, Confluence, Microsoft Teams, Slack, Mattermost, Medium

Leadership, Service & Recognition

- Ambassador: Women in Data Science (WiDS) Ethiopia
- Poster Judge: Deep Learning Indaba
- Team Advisor: Women-focused AI Hackathon
- Reviewer: Africa at NeurIPS, WiML @ NeurIPS, NeurIPS GlobalSouthAI, AGI
- Featured: WIA Philanthropy weekly spotlight, 2026
- Recipient of competitive research visit scholarships and conference grants from CVPR, ICCV, NeurIPS, WiML, Black in AI, Deep Learning Indaba, AGI, BGI, CyFy, ITEC, SIDA, ACWES, GITEX Global, Blockchain Summit, the Cloud and Supercomputing Seminar, Women Elevate, 10 Academy, and IOHK Haskell Training.

Community Engagement

- Volunteer supporting vulnerable mothers and children through providing food and clothes